Hearing Friends in Loud Places: Form-Factor Simulation and Solution Map

Speech Enhancement Benchmark Lab, Round 4 (October 18, 2026)

1. Bottom line

- **The bottleneck is how clean the device's own copy of your friend's voice is, not cancelling the room.** In our open-ear simulation, a wearable array (glasses, neckband) or a tabletop ring adds about **+3.2 to +3.8 dB** of equivalent SNR at the ear, or +4.2 to +5.3 dB when its output is played 20 dB above the direct voice. That matches the **2.2 to 4.4 dB** measured for Nuance Audio hearing glasses, which validates the model.
- **There is a lot of headroom.** If the device's signal were 10 dB cleaner than the room (device SNR 10 dB), the same open-ear delivery would give **+7.3 dB** (at +10 dB playback) to **+11.3 dB** (at +20 dB). At a device SNR of 15 to 20 dB it gives up to about **+14.5 to +16.6 dB**. That is the target-speaker-extraction or remote-mic regime.
- **Delay costs about 0.2 to 0.3 dB of benefit per millisecond.** Keep the device's total delay at **5 ms or less** for open delivery. At 17 ms, glasses fall from +3.8 to +1.0 dB; at 30 ms, everything is roughly zero or negative. Bluetooth-class streaming (about 20 ms or more) erases most of a remote mic's advantage.
- **What doesn't work:** a plain amplifier (one mic, no directionality) makes things worse (−0.5 dB). Neither "averaging" babble (stationary-noise suppression) nor adding an 8 ms single-channel post-filter helps. Open-air active cancellation can't touch speech-band babble.
- **What does:** get the cleanest possible capture (mic close to the talker, a big array, voice-aware extraction), deliver it within 5 ms, and play it close to the ear so it doesn't raise the room level.

2. Simulation method

- **Scenes:** 21 reverberant bar scenes (RT60 0.55 to 0.85 s, 10 to 14 m rooms), each at input SNR −5 and 0 dB, giving 42 cells.
  - The friend talks from 1.0 m across a table.
  - Noise comes from 8 talkers at 1.6 to 5 m, plus 2 loudspeakers playing recorded crowd babble (MS-SNSD).
  - Speech is from CMU ARCTIC and LibriSpeech, with image-method acoustics (pyroomacoustics).
- **Capture options:** 6-mic glasses frame; 6-mic neckband (20 cm below the ears); 8-mic tabletop ring (0.5 m ahead); a phone or clip mic on the friend; a 1-mic ear-level amplifier.
  - The arrays use a fixed superdirective beamformer with near-field steering to the friend, implemented as a causal FIR (2 or 8 ms).
  - An optional 8 ms streaming Wiener post-filter was also tested.
- **Delivery:** the ear receives the direct sound plus the device output, delayed and played 0, 10 or 20 dB above the direct voice. Open-ear delivery leaves the noise at full level. A 25 dB isolating "in-ear" case is included for comparison.
  - An **ideal device** that emits the perfectly clean voice sets the physics ceiling.
- **Metric:** ESTOI, measured against the friend's early (direct + 50 ms) sound at the ear, converted to an **equivalent SNR benefit**. That is how many dB quieter the room would need to be, unaided, to understand equally well. This is the same unit hearing devices are rated in. 95% CIs come from scene-level variation.

- **Capture sweep:** a separate sweep replaced the beamformer with a device signal of controlled cleanliness (device SNR 0 to 30 dB). This maps *any* front end (beamformer, causal TSE, table hub) onto the listener's benefit.
- **Limitations:** no head or torso acoustics, no acoustic feedback (which limits real open-ear gain), no Lombard speech, ESTOI rather than human listeners, and a static talker.

3. Results

Benefit by form factor (open ear, shortest realistic delay)

Form factor	Delay	+10 dB playback	+20 dB playback
Plain amplifier (1 mic at ear)	3 ms	−0.5 dB	−0.2 dB
Neckband, 6 mics	3 ms	+3.2 dB	+4.2 dB
Tabletop, 8-mic ring	4.5 ms	+3.6 dB	+5.3 dB
Glasses, 6 mics	3 ms	+3.8 dB	+4.7 dB
Phone/clip mic at friend	7 ms	+4.9 dB	+6.2 dB
Ideal clean voice (ceiling)	1 ms	+9.8 dB	+19.7 dB

All CIs are within ±0.35 dB.

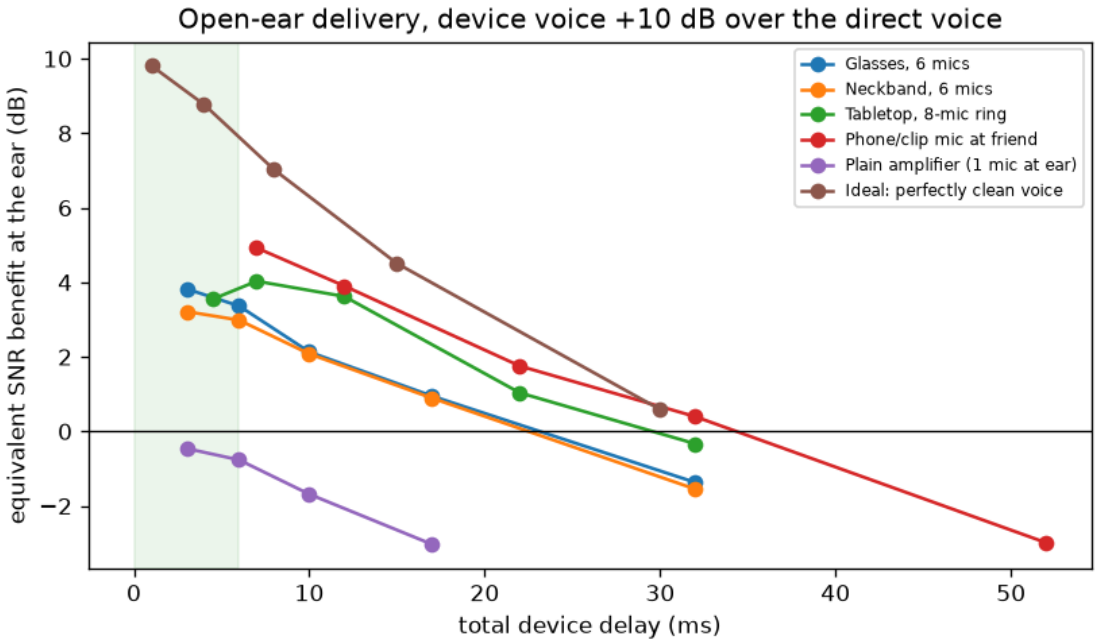


Figure 1: Benefit versus delay

Delay.

- Glasses: +3.8 dB at 3 ms, +2.1 at 10 ms, +1.0 at 17 ms, −1.4 at 32 ms.
- Phone mic: +4.9 dB at 7 ms, +1.75 at 22 ms, −3.0 at 52 ms.
- Ideal voice: +9.8 dB at 1 ms, +4.5 at 15 ms, +0.6 at 30 ms.
- The delayed copy blurs the speech envelope and comb-filters the direct sound. That is the same physics behind the hearing-aid literature's 5 to 10 ms open-fit limit.

Device cleanliness is the lever.

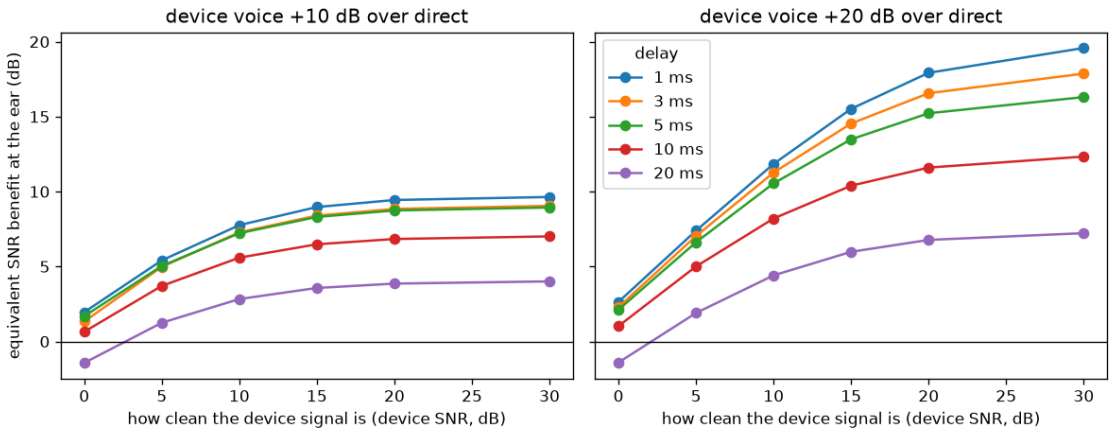


Figure 2: What a cleaner device signal buys

- At 3 ms and +20 dB playback, the benefit is +2.3 dB at device SNR 0, +7.0 at 5, +11.3 at 10, +14.5 at 15, and +16.6 at 20.
- Our fixed beamformers only raise the device SNR about **1.3 to 2.6 dB** above the ear's own SNR in diffuse babble. The phone at the friend raises it about **5.3 dB**.
- Round 3 showed offline target-speaker extraction adding about +10 dB SI-SDR on unseen talkers. UW's causal TSE gets 7 dB at 12 to 18 ms. **A low-latency voice-aware front end is therefore worth roughly 2 to 3 times the benefit of today's glasses.**

Post-filter:

- An 8 ms Wiener post-filter *lowered* the benefit for glasses (+3.8 to +2.6 dB) and the neckband (+3.2 to +2.1 dB).
- It added nothing at the tabletop (+3.55 to +3.65 dB).
- It costs delay and doesn't remove babble.

In-ear comparison: isolating the ear adds only about 1 dB with these noisy device signals. With the ideal clean voice it adds +27 dB at 1 ms. Isolation only pays off when the capture is clean.

4. Predictive traffic experiment

- **Scene:** a street café, with 30 sentences × SNR −5/0/+5 dB and 1 to 3 cars passing per clip.
  - Cars are ESC-50 engine textures shaped by the physical pass-by law $L(t) \propto 1/(d0^2 + v^2(t-t0)^2)$.
  - Every variant uses the same 8 ms streaming Wiener filter.

Noise estimate	ΔSI-SDR	ΔESTOI
Blind minimum-statistics tracker (usual)	+0.8 dB	+0.011
Blind fast tracker (0.3 s)	+0.7 dB	−0.027
Predictive (up-street sensor forecasts each car's rise; ±2 dB error)	**+3.7 dB**	+0.023
Oracle noise (upper bound)	+5.2 dB	+0.111

Predicting the pass-by **quadruples noise removal** (+0.8 → +3.7 dB SI-SDR), because blind trackers lag each car by about a second. With a simple low-latency gain rule it barely moves intelligibility (+0.023 ESTOI). It's useful for comfort, and as a feed into a smarter suppressor or a gain-riding voice boost, but it isn't a hearing solution on its own.

"Averaging" babble (stationary-noise suppression) was measured in Round 3 on recorded café babble: +0.9 dB SI-SDR and +0.009 ESTOI. It's real but tiny, as theory predicts, because babble's average spectrum is speech's spectrum.

5. Solution map: seven approaches at different levels

1. Table hub + neckband ("virtual headphones without headphones")

- **How it works:** a centerpiece mic array runs beamforming plus voice-aware extraction for each tablemate, with enrolment-free speaker clustering. It streams each voice over a ≤5 ms proprietary radio link to a neckband with near-ear speakers 10 to 15 cm from each ear, played a few dB above the local noise.
- **Expected benefit: +7 to 11 dB** if the hub reaches device SNR ≥10 dB and ≤5 ms (sim). Today's parts give +3.5 to 5 dB.
- **Maturity:** components ship (Roger/Orsana table mics, neck speakers, LE Audio); the integration doesn't. Patent TableConverse (US 12,354,619) is nearby prior art.
- **Key risk:** radio latency, feedback, privacy.

2. Wireless look-ahead relay ("faster than sound")

- **How it works:** a table or venue node radios the sound before it arrives acoustically (about 3 ms per metre of head start). That buys compute time for heavier extraction, or a reference for cancellation (MUTE, SIGCOMM 2018: open-ear cancellation 0 to 4 kHz for a single source).
- **Expected benefit:** turns the 5 ms budget into about 8 to 15 ms of usable compute for distant noise sources.
- **Maturity:** research.

3. Large-aperture wearable array with gaze or head steering (glasses / neckband)

- **How it works:** 6 to 16 mics, steered toward where you look (Kidd's visually guided hearing aid) or face, plus a tiny causal neural spatial filter (GCFSnet, 5.4 ms, +1.3 dB over the classical beamformer).
- **Expected benefit: +4 to 7 dB** (sim plus literature: Varibel 5.3 to 6.3, Widrow necklace >10 dB measured in 2003).
- **Maturity:** Nuance (2.2 to 4.4 dB) and Meta (no data) ship; a bigger array plus AI is open space.

4. Predictive environmental modeling (traffic, music, HVAC)

- **How it works:** forecast each noise event's rise from up-street or venue sensors (car pass-by, bus, DJ drop). Pre-adjust suppression and voice boost before the noise peaks; use order-tracking for engine harmonics.
- **Expected benefit: +3 dB less noise** in our traffic sim (vs +0.8 blind). It's a comfort feature, with a small intelligibility gain unless paired with a better suppressor.
- **Maturity:** no published hearing-device work (gap). The closest is Oticon's motion-intent sensors.

5. "Duck the DJ" and venue integration

- **How it works:** the house music is the one loud noise available *electronically before it plays*. Use it as a perfect cancellation reference for bass at chair-backs or neckbands (quiet zones of 7 to 34 cm below 500 Hz). Or zone and duck venue speakers near tables that request it.
- **Expected benefit:** 5 to 10 dB less music bass locally (inferred from headrest ANC results); whole-venue absorption gives up to about 10 dB of reverberant reduction.
- **Maturity:** speculative as a product; the ANC pieces are proven in cars and headrests.

6. Seat, booth or headrest local ANC plus near-ear speech

- **How it works:** head-tracked headrest cancellation (Elliott 2018: about 20 dB at 300 Hz to 1 kHz in the lab; Silentium claims about 10 dBA), or booth-aperture cancellation (Lam 2020 window ANC: 7.5 to 10 dBA). Add seat-back speakers for boosted speech.
- **Expected benefit:** good on rumble, hum and music, little on chatter; only when people lean back.
- **Maturity:** car and aircraft seats are near production; a restaurant chair is speculative.

7. Directional delivery: ultrasonic or speaker-array "sound spotlights" from a lamp or centerpiece

- **How it works:** parametric beams (Holosonics, 85 to 95 dB max, about 0.5 m spot at 2 m) or sound-zone arrays (about 15 dB of bright/dark contrast) put the boosted voice at one head without filling the room, which avoids the Lombard spiral.
- **Expected benefit:** about +5 to 10 dB of speech at the target head, weak bass, needs head tracking.
- **Maturity:** hardware ships; table use is unproven. Check ultrasound exposure limits.

Not worth pursuing:

- Open-air ANC of babble at speech frequencies (the quiet zone is about 3 cm at 1 kHz).
- Masking noise (it lowers the SNR).
- One-mic amplifiers.
- Single-mic neural denoisers at $\leq$16 ms (Round 3).
- Bluetooth-latency phone streaming to open ears.

6. Competitive landscape

Hearing glasses

- Nuance Audio (6 mics, measured +2.2 to 4.4 dB, $699): Nuance Audio Plus launched Sep 29, 2026, from $829.
- Legato raised $12M in Aug 2026.
- Meta's Conversation Focus (no published dB; capped free use).

DNN hearing aids

- Phonak Sphere Infinio: +5.9 dB vs open ear (vendor), in-ear, prescription.
- Phonak Roger On Sphere: an AI table/remote mic shipping mid-Nov 2026.

Remote / table mics: Roger, Orsana. Remote mics measured +14 to 22.7 dB, but listeners wear receivers or earbuds.

Research leaders: UW (Target Speech Hearing 7 dB causal; Sound Bubble), Oldenburg (GCFSnet), Boston University (Kidd, gaze steering), MUTE (UIUC).

White space:

- An earbud-free system that uses a table node's clean capture plus $\leq$5 ms near-ear delivery, sold to people with normal hearing.
- Nobody publishes conversation-benefit numbers for neck-worn speakers.

7. Recommended next experiments

1. **Causal TSE at $\leq$5 ms on a table hub:** distil the Round-3 TSE into a streaming model, then feed it into the capture sweep. This tests whether a device SNR of 10 dB or more is reachable.
2. **Head and torso model plus feedback:** add HRTFs and speaker-to-mic feedback paths for the neckband and glasses, to find the maximum stable playback gain (we assumed up to +20 dB).
3. **Real recordings:** an 8-mic dev board (e.g. ReSpeaker or miniDSP UMA-8) at a real Omaha restaurant, run through the same pipeline, plus a 12-person listening test with the existing app.
4. **Predictive suppressor v2:** pair the pass-by forecast with a neural suppressor and gain-riding voice boost, and measure ESTOI.
5. **"Duck the DJ" bench test:** headrest-scale ANC with an electronic music reference, measured on a dummy head.

Appendices A and B: full research memos with sources.

Appendix A: Open-air noise control research

*Research brief for the "hear your friends in loud bars / restaurants / streets" product. Compiled 2026-10-18. Every number below comes from a source I opened unless it is marked **[theory]** (standard acoustics I worked out myself), **[excerpt]** (seen only in a search snippet or vendor copy, not checked against the full text), or **[speculative]**.*

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0. Bottom line

1. **Physics caps open-air ANC for speech-band noise.** A local "zone of quiet" (≥10 dB of reduction) is only about $\lambda/10$ across: about **34 cm at 100 Hz, 6.8 cm at 500 Hz, 3.4 cm at 1 kHz, 1.7 cm at 2 kHz** [theory; the 3.4 cm @ 1 kHz figure is stated in Elliott et al. 2018 and Lam et al. 2020]. Speech intelligibility sits mostly between 500 Hz and 4 kHz. To quiet a 1–4 kHz zone at someone's ear without earbuds, you need something right next to the ear, plus head tracking, plus lookahead.
2. **Restaurant babble is close to the worst case for ANC.** It comes from many uncorrelated sources in a diffuse field. One upstream reference mic cannot predict it, and deep-learning noise prediction loses performance quickly beyond about 6 ms of lookahead, even in simulation (Zhang, Pandey & Wang 2023). Published ANC results on real multi-source babble at a person's head, without earbuds, are close to nil.
3. **Open-air ANC is real, but only below about 1 kHz and in contained geometries.** Lab results: about 20 dB at a headrest from 300 Hz to 1 kHz with head tracking (Elliott 2018). About 10 dB across a whole room through an open window, 500 Hz–1 kHz, using 24 speakers (Lam 2020). In production: about 3 dB of road noise in cars (Hyundai/Harman RANC 2020) and "up to 10 dBA, 30–1000 Hz" from Silentium's headrest "Quiet Bubble". These numbers suit bass and rumble (bar music, traffic, HVAC), not chatter from the next table.
4. **The lever that works for "hearing friends" is raising SNR, not removing noise.** That means picking up the talker close to the mouth (table or worn mic arrays) and delivering their voice to the listener's ear, as Phonak Roger, Orsana, TableTalk and hearing glasses do. Remote mics typically take listeners from about 27% to 83% speech understanding in noise [excerpt, Phonak]. The unsolved part is delivering that voice **without earbuds** and **without making the room louder**, since amplification feeds the Lombard spiral. Near-field neck or chair speakers, head-tracked beamforming, and parametric beams are the candidates.
5. **My recommendation for form factor:** a **shared table hub** (beamforming mic array, AI speech separation, voice-lift) paired with **per-person near-ear delivery** (neckband near-field array, chair-back or headrest array, or open-ear glasses). Bass-band ANC in a chair-back or booth is optional. The ANC part of that stack is incremental. The SNR-delivery part is the product.

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1. The numbers that set the constraints

Quantity	Value	Source
Zone of quiet (≥10 dB), diffuse field	$\approx \lambda/10$ diameter; 3.4 cm at 1 kHz	Elliott, Jung & Cheer 2018, Sci Rep https://pmc.ncbi.nlm.nih.gov/articles/PMC587639
$\lambda/10$ by frequency	100 Hz → 34 cm; 250 Hz → 13.6 cm; 500 Hz → 6.8 cm; 2 kHz → 1.7 cm; 4 kHz → 0.85 cm	[theory] c = 340 m/s
Head movement that breaks a fixed ZoQ	~5 cm	Elliott 2018
Headphone ANC timing budget	≈ 30 μs (ref mic to speaker < 1 cm)	Shen et al., MUTE, SIGCOMM 2018 https://sens.epfl.ch/documents/papers/sigcomm18
Lookahead per metre of path difference	≈ 3 ms/m (sound vs. RF)	MUTE 2018

Quantity	Value	Source
Causality rule	T_ANC ≤ T_primary − T_secondary	Zhang, Pandey & Wang 2023 https://haozhang6720.github.io/doc/papers/ANC_
Car RANC budget	noise path ≈ 9 ms, DSP 2 ms	Hyundai https://www.hyundaimotorgroup.com/en/story/C
Restaurant occupied levels	45–82 dBA in 10 eating places (Hodgson 2007); 63–77 dBA LAeq in 5 Omaha restaurants (Paine, UNL); instantaneous 58–97.5 dBA in Orlando (CDC summary)	https://pubmed.ncbi.nlm.nih.gov/17471719 ; https://digitalcommons.unl.edu/cgi/viewcontent.cg ; https://stacks.cdc.gov/view/cdc/223002/cdc_2230
Lombard slope	speech level rises 0.5–0.7 dB per dB of noise above ~45 dB	Rindel 2012 https://odeon.dk/pdf/C116-BNAM_2012_Rindel_29.pdf
Disturbance / vocal-effort onset	52.2 dBA (disturbance), 57.3 dBA (vocal effort rises)	Bottalico 2018 via UIUC https://www.shs.illinois.edu/2019/07
Rindel "acoustic capacity"	past capacity, ambient > ~71 dB means communication is "insufficient"	Rindel EuroNoise 2015 https://odeon.dk/pdf/C127-Keynote%20EuroNoise%202015%20Rindel.pdf

Implication [theory]: in a 75–80 dBA bar the Lombard effect pushes conversational speech up, but SNR at a listener 1 m away still ends up around −5 to +5 dB. Normal-hearing listeners need roughly −6 to −8 dB SNR for 50% sentence recognition in babble. (Mild-to-moderate hearing-loss adults in the Nuance study scored −8.3 dB unaided on a matrix test; that is a different population.) Hearing-impaired listeners need far more: 15–25 dB SNR for severe-to-profound loss [excerpt, Phonak/Baquis]. Each 3–6 dB of SNR gain is worth a lot of intelligibility on the steep part of that curve.

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2. Open-air / free-field ANC

2a. Local ANC (headrests, seats): virtual sensing and the remote-microphone technique

- **How it works.** Loudspeakers in a headrest or seat cancel noise at "virtual" ear positions. Pressure at the ear is estimated from fixed monitoring mics (the remote-microphone technique, RMT). Optional head tracking switches the ear-position filters as the head moves.
- **Best measured result. About 20 dB broadband, 300 Hz–1 kHz, at the ears of a dummy head.** Setup: 2 headrest speakers, 4 monitoring mics 18–27 cm from the head, Kinect head tracking across 20 grid positions 5 cm apart, 3 kHz sampling, total plant delay about 1.3 ms. Without head tracking, about 10 dB only up to about 600 Hz once the head moved 14 cm. Front-arriving noise needed a 0.7 ms modelling delay to stay causal. (Elliott, Jung & Cheer, Sci Rep 2018, https://pmc.ncbi.nlm.nih.gov/articles/PMC5876390.) Important caveat: the reference signal came straight from the noise source, with a single source in an anechoic room. The authors call this idealised.
- **Aircraft-seat virtual sensing (2023).** A functional-link neural network with linear pressure prediction grew the quiet zone from 10 × 40 cm to **25 × 50 cm**. Some harmonics were reduced by about 20 dB, for tonal engine noise (https://link.springer.com/article/10.1007/s42417-023-01090-5).
- **EU business-jet seat ANC target.** 15 dB, with at least 10 dB at the ~110 Hz engine tones, in a ≤ 9 kg system (CORDIS H2020 717089, https://www.cordis.europa.eu/project/id/717089/reporting/es).
- **Commercial headrest.** Silentium "Quiet Bubble": **"up to 10 dBA, 30–1000 Hz"** from a speaker array in a headrest (vendor claim, https://www.silentium.com/technology-2/quiet-bubble).
- **Maturity.** Lab to pre-production for headrests. In cars, cabin-wide low-frequency ANC is in production (below).
- **Constraints.** Noise must be correlated with a reference, so it works for tonal or structure-borne noise and fails for babble. Only below about 1 kHz. The head must stay within about 5 cm of the trained positions, or be tracked.

- **Form factors.** Chair-back or headrest, high-back restaurant booth, car seat. **Realistic benefit in a restaurant chair: about 5–10 dB of reduction in music bass and HVAC hum below about 500 Hz at the seated head; little to none on speech-band babble** [inferred from the results above].

2b. Global or aperture ANC (windows, openings)

- **Lam, Shi, Gan, Elliott, Nishimura, Sci Rep 2020** (https://pmc.ncbi.nlm.nih.gov/articles/PMC7347542). 24 small speakers (4.5 cm) spaced 12.5 cm apart across a 0.45 × 0.93 m opening, 1 reference mic, a fixed filter trained beforehand (no error mics while running), FPGA control. Results: **7.5–10.1 dBA room-averaged reduction** for highway, MRT train and aircraft noise in the 100–1000 Hz band; **about 10 dB above 500 Hz; nothing below 300 Hz** because the speakers are small. For comparison, closing the window passively gave 12.4–13.9 dB.
- **Follow-up listening test** (anti-noise window, 2023): https://arxiv.org/pdf/2307.05533.
- **Survey table in Lam et al. 2021, "Ten questions concerning ANC in the built environment"** (https://www.sciencedirect.com/science/article/pii/S0360132321003322): window results of 4.6–10.5 dB in the 100–700 Hz band; about 15.5 dB below 300 Hz (Carme); about 3 dB on a real aircraft pass-by. It warns about "spillover", meaning noise gets louder outside the controlled zone.
- **Form factors.** Booth openings, a semi-enclosed pod or canopy aperture, a patio screen. **Speculative:** a booth or pod whose open side is an "active aperture" could take about 5–10 dB off the 300 Hz–1 kHz band of incoming sound. That requires sound entering through a defined opening, which an open table does not have.

2c. Automotive production systems (the most mature open-air ANC)

- **Hyundai / Harman RANC** (production from 2020 on the Genesis GV80). Accelerometers on the suspension supply a reference that leads the sound. DSP takes 2 ms against a 9 ms noise path. **Road noise reduced by about 3 dB ("up to 50%")** (https://news.harman.com/releases/releases-20200130 ; https://www.analog.com/en/signals/articles/hyundai-noise-cancellation-technology.html).
- **Bose QuietComfort RNC** (2019): same idea, using accelerometers and cabin mics (https://assets.bosecreative.com/m/3a
Introduces-).
- **Lesson.** Even with structure-borne references that arrive early, a closed cabin and a huge budget, production gains are a few dB, below a few hundred Hz.

2d. ANC for open-plan offices

- Real ANC deployments in offices are essentially headphones. Field studies note that ANC headphones handle noise below 1 kHz well but cancel speech less well, because speech is transient (Frontiers 2022, https://www.frontiersin.org/journals/built-environment/articles/10.3389/fbuil.2022.962462/full).
- "Noise-cancelling office chair" work exists (Treyer et al., ICSV 2016, cited in Elliott 2018). I did not find published dB figures for it.
- Office acoustics is dominated by absorption and masking (section 5).

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3. Predictive / lookahead ANC

3a. Wireless-forwarded reference (MUTE; Shen, Roy, Guan, Hassanieh, Roy Choudhury, SIGCOMM 2018)

- **How it works.** An IoT relay mic placed toward the noise source forwards audio over a 900 MHz analog FM link. RF arrives about 3 ms per metre ahead of the sound. The lookahead lets the system run longer non-causal inverse filters, and switch filter profiles when speech starts or stops.
- **Result** (https://sens.epfl.ch/documents/papers/sigcomm18mute.pdf). With an **open ear**, MUTE cancelled across **0–4 kHz**; Bose QC35 ANC alone covers only up to about 1 kHz. Below 1 kHz MUTE beat Bose ANC by 6.7 dB. It was only 0.9 dB worse than Bose's ANC plus passive earcups (about −15 dB average). Profile switching added about 3 dB on intermittent speech. The 4 kHz ceiling came from the DSP (TMS320C6713 at 8 kHz sampling).

- **Caveats.** A single dominant source. The relay must sit between source and listener. The speaker sat right next to an error mic at the ear position of a head model, so the "hollow ear device" was never actually built. The paper's own "Personal TableTop" variant (fig. 10a) is very close to this product idea.
- **WINC (2023)** extends this to multiple sources using a network of IoT mics (https://dl.acm.org/doi/fullHtml/10.1145/3583 I read only the summary.
- **Form factors.** A tabletop hub as the relay plus an open-ear device, or "smart noise" relays fixed to the noise sources: a speaker system in a bar could broadcast its own feed. **Speculative but promising:** a venue's music feed is a *perfect reference*, available electronically before it plays. Cancelling the house music at a chair-back or neckband is the one bar noise that is fully known ahead of time.

3b. Deep-learning ANC (Zhang & Wang)

- **Deep ANC (Neural Networks 2021).** A convolutional recurrent network (CRN) predicts the anti-noise and handles loudspeaker nonlinearity, in simulation (https://pmc.ncbi.nlm.nih.gov/articles/PMC8328877/).
- **Low-latency ARN (IEEE/ACM TASLP 2023)** (https://haozhang6720.github.io/doc/papers/ANC_journal_2023.pdf Time-domain attentive RNN with 4 ms frames. With revised overlap-add plus 2 ms of prediction, **algorithmic latency was 0 ms at an average NMSE of −11.2 dB**, only 0.47 dB worse than non-causal. Predicting **babble 15 ms ahead cost only about 0.95 dB** (frame-based model). With 4 ms frames, predicting more than about 6 ms ahead caused a "considerable drop". The model was also tested on DEMAND recordings, including PRESTO (a university restaurant) and SPSQUARE (a town square).
- **Caveat.** Everything was simulated: a single reference, a single loudspeaker, one error point, a $3 \times 4 \times 2$ m room with T60 = 0.2 s. These are point-cancellation figures, not quiet zones. The model has about 15.9 M parameters and about 5.2 GMAC per 3 s of audio.
- **Use for the product.** Mainly as a better controller inside a headrest, chair or neckband, and to stretch lookahead by a few ms. It does not change the $\lambda/10$ limit.

3c. Traffic pass-by and periodic noise

- Periodic noise (engines, HVAC, fans, music beats) is predictable, so causality is not the limit. Narrowband ANC works best on this class, and it is the only class where long prediction is reliable (Zhang 2023 discussion; Lam 2021).
- Traffic pass-by: the anti-noise-window studies show about 7.5–8.7 dB in the 100 Hz–1 kHz band on recorded traffic and aircraft (Lam 2020). Lam 2021 lists about 3 dB on real aircraft pass-bys. I found **no published "predict the next pass-by event" ANC at a person** in open air; the search turned up only traffic-noise monitoring work. Treat that idea as [**speculative**].

3d. Babble: why ANC fails

- **Physics [theory plus sources].** ANC is feedforward: it can only cancel the part of the noise that is *coherent* with a reference. In a diffuse field, coherence between two points falls off roughly like sinc(kr). At 1 kHz it is essentially gone beyond about 17 cm [theory]. Babble also comes from dozens of independent talkers, each needing its own reference (MUTE limitations section; WINC).
- **Perception.** Babble masking is part energetic, part informational. Babble of 2–4 talkers is the most damaging (informational masking peaks). With more talkers it turns into energetic, noise-like masking (https://pmc.ncbi.nlm.nih.gov/articles/PMC3631264/ ; https://doi.org/10.1121/1.1345696 ; open-office study, about 4 talkers: https://arxiv.org/pdf/2304.12817). Partial ANC that thins a crowd into a few audible nearby talkers could **raise** informational masking. That is a non-obvious risk [inferred].
- **Results.** No published open-air (no-earbud) ANC result on real restaurant babble at a moving head that I could find. The closest are Deep ANC simulations (3b) and MUTE's single-talker cancellation (3a).

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4. Personal sound zones and directional delivery

4a. Loudspeaker-array sound zones (acoustic contrast control, pressure matching)

- **How it works.** An array is driven so that energy is maximised in a "bright" zone and minimised in a "dark" zone (ACC), or so that target fields are matched (PM).

- **Measured contrast.** About **15 dB in practical set-ups**, which the authors say is "typically not high enough to satisfy the people inside the zones" (Lee et al., https://arxiv.org/pdf/1911.10016). In cars, 15–25 dB between front and rear seats (Cheer & Elliott), 20–40 dB below 1 kHz with 16 speakers (Choi), lower when measured with a dummy head (ICA 2019, https://pub.dega-akustik.de/ICA2019/data/articles/000245.pdf). 2025: relaxed-reverberation PM added more than 3 dB of contrast at 0–2 kHz (JASA, https://pubs.aip.org/asa/jasa/article/157/2/778/3333907). A neural PSZ testbed used 30 speakers on a 1.68 m-radius circle for 0.4 × 0.4 m zones 1 m apart (https://arxiv.org/html/2512.10375v1).
- **Neckband version.** Jeon & Choi (KAIST, JASA 2020): **3 speakers next to each ear on a neckband**, two second-order IIR filters, a broadside differential pattern that boosts near-field sound at the ear relative to the far field. Measured in simulation and experiments (I saw the abstract only; no dB numbers) (https://pubmed.ncbi.nlm.nih.gov/33379876).
- **Head-tracked beamforming and crosstalk cancellation (commercial).** Audioscenic Amphi: a camera tracks the head and two beams go to the ears. Shipped in the Razer Leviathan V2 Pro (CES 2023) and Acer SpatialLabs View (2024) (https://www.audioscenic.com/ ; http://audioscenic.com/about). It is built for spatial audio, not noise, but it proves desktop-scale ear-targeted beams are manufacturable.
- **Constraint.** Sound zones control *where your audio goes*. They do not remove ambient noise. For this product, a sound zone is the **delivery** mechanism for boosted speech, so neighbours are not disturbed and the Lombard spiral is not fed.

4b. Parametric / ultrasonic directional speakers

- **Holosonics Audio Spotlight.** A film transducer with a ~60 kHz carrier. AS-168iX: **85 dB SPL @ 1 kHz max; 5° beam; covers 0.5 × 1 m at 2 m**. AS-16iX: 90 dB. AS-24iX: 95 dB (https://www.holosonics.com/dakota-audio ; spec sheet https://www.touchwindow.com/mm5/graphics/00000001/holosonic Vendors position them for "quiet environments". Weak bass, and at 85–90 dB max they barely clear a 75–80 dBA bar.
- **Noveto SoundBeamer / i3DS.** Ultrasonic beams with a 3D camera form "two pockets of sound outside the ears". SDK spec: **85 ± 5 dB SPL, up to 2 m, 750 Hz–20 kHz (±6 dB), −20 dB at 1 m to the listener's side, −15 dB behind, 256 actuators, ear tracking at 2–30 fps** (https://www.noveto.com/licensing). Status: soundbars shown at CES 2024 (PR). **PitchBook lists Noveto as "Out of Business / Liquidation"** (https://pitchbook.com/profiles/company/127676-35) [excerpt; CB Insights still shows "alive"]. Treat it as defunct.
- **Single-channel multi-carrier parametric sound zones** (Zhuang et al., arXiv 2504.17440 / ICASSP 2026). Different audio is modulated onto several ultrasonic carriers so one transducer behaves like a multi-channel array. **Simulation only** (https://arxiv.org/abs/2504.17440).
- **Form factors.** Overhead lamp or pendant above a seat, centerpiece aimed upward, table edge. **Realistic:** at ~85–90 dB max, a parametric beam could add speech at about +5 to +10 dB over a 75–80 dBA background at one head position in a beam roughly 0.5 m wide at 2 m. Low-frequency speech energy (below about 500 Hz) is weak, and beams reflect off tables and faces [inferred]. I did not research ultrasound exposure limits; the risk is flagged in section 7.

4c. Beam-steered speech reinforcement at a table ("voice lift")

- **TableTalk Labs.** A patented wireless voice-lift device for noise above 75 dB, with directional mics and speakers that boost high frequencies (https://www.tabletalklabs.com/) [vendor copy, no numbers].
- **US Patent 12,354,619 "TableConverse"** (H. Perritt, granted 8 Jul 2025). Beamforming mics in the table perimeter (100 Hz–8 kHz), ML noise suppression, parametric speakers mounted overhead or in the table with laser seat-positioning dots, latency under 15 ms, conversation-group routing, and **bone-conduction transducers in chair headrests and seatbacks** as an alternative. Claims only, no measurements (https://patents.justia.com/patent/12354619). This is relevant prior art for any table-hub plus chair delivery design.
- **Remote/table mics plus worn receivers (the benchmark to beat).** Phonak Roger Table Mic II / Roger Select (MultiBeam, 6 beams; Roger On Sphere adds about 3 dB SNR over Roger On 3 at −15 dB SNR / 80 dB noise). Phonak field studies: 83% vs 27% speech understanding in 60 dBA noise with Roger plus hearing aids versus hearing aids alone; up to 61% improvement in simulated restaurant noise [excerpts from

Phonak documents: https://www.phonakpro.com/content/dam/phonakpro/gc_hq/en/resources/evidence/field_studies/d
; https://www.phonak.com/content/dam/phonak/en/evidence-library/field-studies/PH_Insight_ExpandYourWorldWithR
2933-02.pdf]. AudioTelligence Orsana: a 6 × 6 cm tabletop unit using blind source separation, LE Audio/Auracast to earbuds, crowdfunded in 2023 (https://audioxpress.com/news/audiotelligence-announces-orsana-le-audio-assistive-listening-device). These all deliver to the ear through earbuds or hearing aids. The open gap is earbud-free delivery.

- **Open-ear hearing glasses (no earbuds, closest shipping competitor).** EssilorLuxottica Nuance Audio: FDA-cleared OTC (Feb 2025), beamforming mics, open-ear speakers in the temples. A 2026 study gave **SRT50 of −10.55 dB SNR aided vs −8.34 dB unaided, about 2.2 dB better**, in mild-to-moderate hearing loss (Frontiers 2026, https://www.frontiersin.org/journals/audiology-and-otology/articles/10.3389/fauot.2026.1822238/full). The company claims "up to 29% better speech understanding" (Jan 2026 PR). Nuance Audio Plus launched 29 Sep 2026. Legato (US) came out of stealth Aug 2026 with $12M for AI voice-isolating open-ear hearing glasses (TechCrunch, https://techcrunch.com/2026/08/26/hearing-tech-startup-legato-emerges-from-stealth-with-12m-and-a-peek-at-its-ai-hearing-glasses/).

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5. Masking, psychoacoustics and passive treatment

- **Sound masking.** Steady broadband noise covers intelligible speech by raising the floor. It is used in offices for privacy, typically 45–51 dBA (Acoustics 2002, https://www.acoustics.asn.au/conference_proceedings/AAS2002/AAS200
In a bar this works against the goal: adding noise lowers SNR. The only relevant use is *masking neighbouring tables' intelligible speech*, which reduces informational masking and distraction, at a cost in energetic masking. It is a nuanced lever, and only for speech that is already quieter than your companions' voices [inferred].
- **Informational vs energetic masking.** Covered in 3d. The practical design rule: get the target voice spatially and spectrally distinct and a few dB louder, and informational masking falls away quickly. Different-sex and spatially separated talkers are much easier to follow (Brungart 2001).
- **Lombard feedback.** Any system that radiates amplified speech into the room raises the noise floor for everyone, and diners raise their voices by 0.5–0.7 dB for each dB. Delivery must be near-field or beamed (Rindel 2012).
- **Passive room treatment.** An absorptive ceiling (NRC $\geq$ 0.8) "can reduce reverberant noise levels by 10 dB" (manufacturer white paper, https://www.tombarrow.com/images/literature/silence-is-golden-optimizing-the-acoustic-environment.pdf) [vendor claim]. Rindel's capacity model (Rindel 2012, 2015) states that the occupied noise level depends on absorption per person, and that ceiling treatment alone may not be enough. Treatment works on the reverberant field and helps everyone, but it is a venue sale, not a personal device. It also reduces the Lombard spiral, so gains compound.
- **Acoustic metamaterials.** BU ultra-open silencer: about 60% open area, **94% transmitted-energy reduction (about 12 dB) in a narrow band around 460 Hz**, in a duct (Ghaffarivardavagh et al., Phys Rev B 2019, https://link.aps.org/doi/10.1103/PhysRevB.99.024302 ; band from https://www.sciencedirect.com/science/article/abs/pii/S0003682X1930845X). It is narrowband and needs a defined aperture. Possible use: ventilated booth walls or canopy panels; little use at an open table.
- **Booths and canopies.** I found no peer-reviewed dB figures for restaurant booths or table canopies. **[theory]:** a high-backed booth or canopy mostly blocks the direct path from nearby tables and adds absorption. A plausible gain is a few dB for sound arriving at head height from the side; the reverberant field is unaffected.

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6. Form-factor map: what each approach can realistically do

Form factor	Approaches that fit	Realistic benefit for the user	Maturity	Notes
Tabletop hub / centerpiece	Beamforming mic array plus AI speech separation (pickup); MUTE-style lookahead relay; voice lift	**Pickup SNR gain of about 6–15 dB for tablemates' voices** (Roger/Orsana class) [excerpt]; *still needs* an ear-delivery path	Shipping (Roger, Orsana) with earbuds	Best "brain" of the system. Can also receive the venue music feed electronically as a perfect ANC reference [speculative]
Centerpiece / lamp / pendant (directional out)	Parametric beam(s), head-tracked crosstalk cancellation (Audioscenic-style)	Speech raised by roughly **+5 to +10 dB at one head**, beam about 0.5 m at 2 m; 85–95 dB max SPL caps it in loud bars	Holosonics shipping; Noveto defunct	No ANC; weak bass; needs tracking for movement. One pendant per seat
Neck-worn (neckband / collar)	Near-field 3-speaker-per-side personal audio (Jeon & Choi), local ANC with virtual sensing below about 1 kHz, own beamforming mics or a link to the hub	Speech delivered **10–20 cm from the ear** at a modest level (low spill). ANC: maybe **5–10 dB below about 500 Hz** at the ear [speculative; $\lambda/10$ at 500 Hz is 6.8 cm and the neckband is about 15 cm from the ear]	Research prototypes; neck speakers exist as consumer products	**My top pick** for earbud-free delivery on a street or at a bar. Follows the user. No head tracking needed because the device moves with the torso; head rotation still shifts the ear
Seat / chair-back / headrest	Local ANC with virtual sensing plus head tracking; sound-zone delivery; bone conduction (TableConverse claim)	ANC: **about 10–20 dB, 300 Hz–1 kHz** (lab, idealised reference); about 10 dBA, 30 Hz–1 kHz (Silentium claim); delivery: high contrast to neighbours	Car/aircraft seats near production; restaurant chairs speculative	Requires leaning back; diners lean forward to talk, which breaks the geometry (TableConverse claims posture issues too)
Booth / pod / canopy	Passive absorption plus barrier; active aperture (Lam-style window ANC) on the open side; metamaterial vents; plus any delivery above	Passive: a few dB of direct-path blocking [theory]; active aperture: about **5–10 dB, 300 Hz–1 kHz** for sound entering through the opening (Lam 2020 analogue)	Window ANC is lab/pilot; booths are passive products	A venue sale. Babble enters diffusely, which lowers active gains
Open-ear glasses	Beamforming mics plus open-ear speakers (Nuance, Legato)	**About 2 dB better SRT** in the measured Nuance study	Shipping, FDA-cleared OTC	Direct competitor for the "without earbuds" promise

Form factor	Approaches that fit	Realistic benefit for the user	Maturity	Notes
Whole venue	Ceiling absorption, acoustic-capacity management, speaker zoning of the music	Up to about 10 dB reverberant reduction (vendor claim) plus a smaller Lombard spiral	Mature	Partnering opportunity, not a device

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7. Creative ideas (all [speculative] unless noted)

1. **"Duck the DJ" ANC.** The house music is the one loud bar noise available electronically and earlier than it reaches you. A venue-integrated hub could push the music feed to chair-backs or neckbands for feedforward cancellation of its bass (below about 500 Hz), where zones are large ($\lambda/10 \approx$ 7–34 cm). Basis: MUTE's "smart noise" variant and the causality analysis. Also cheaper: per-zone EQ or ducking of the venue's own speakers near tables that ask for it.
2. **Table hub plus neckband "virtual headphones".** The hub separates each tablemate's voice (beamforming plus BSS, as in Orsana) and streams it over LE Audio/Auracast to each guest's neckband. The neckband's near-field differential array plays the voice about 15 cm from the ear, about 6–10 dB above the local noise, and spatialises it to the talker's real direction (as in the TableConverse claim). Spill to neighbours stays low, so there is no Lombard escalation. Every block exists today; the integration doesn't. Main risks: latency (keep $<$ 10–15 ms for lip sync and so the direct voice doesn't clash with the delayed copy) and acoustic feedback into the hub mics.
3. **Lamp with a downward ultrasonic beam per seat** (TableConverse claim, Holosonics hardware). Limited by 85–95 dB max SPL and weak low frequencies. Tracking keeps the beam on the head.
4. **High-back "conversation chair".** Headrest ANC for bass (10+ dB, 30 Hz–1 kHz class) plus seat-back near-ear speakers for boosted speech. Works only when people lean back. A posture problem for lively conversation.
5. **Partial-ANC "de-intelligibilizer".** Instead of cancelling babble, make nearby intelligible talkers *less* intelligible: add a few dB of speech-shaped masking only in their direction, or use parametric beams to cut informational masking. It relies on psychoacoustics (IM vs EM). Needs listening tests; it could backfire.
6. **Booth active aperture.** Lam-style distributed small speakers across the open side of a booth (think 12–24 channels). Expect at most about 5–10 dB, 300 Hz–1 kHz, and only for noise that comes in through that opening.

Risks to flag. Ultrasonic exposure from parametric arrays (not researched here; check the occupational limits before any near-head design). Acoustic feedback with voice lift. Privacy, since hubs pick up conversations: MUTE kept its relay analogue for this reason. Medical-device rules if marketed for hearing loss: Nuance went the FDA OTC hearing-aid route.

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8. What I could not confirm

- No peer-reviewed dB numbers for: restaurant booths or canopies, the TableTalk device, the TableConverse system (claims only), noise-cancelling office chairs (Treyer 2016), the WINC multi-source results (read the summary only), or the Jeon & Choi neckband contrast (abstract only).
- Noveto status: PitchBook says liquidated, CB Insights says alive. I trust PitchBook more (more recent), but did not verify further.
- Phonak Roger figures come from manufacturer field-study excerpts, not from full texts I read.
- No published open-air ANC at a person on real multi-talker restaurant babble. I believe this is a real gap in the literature, not a gap in my search, but I cannot prove a negative.

9. Key sources (all opened unless marked)

- Elliott, Jung, Cheer 2018, head-tracked headrest ANC — https://pmc.ncbi.nlm.nih.gov/articles/PMC5876390
- Lam, Shi, Gan, Elliott, Nishimura 2020, open-window ANC — https://pmc.ncbi.nlm.nih.gov/articles/PMC7347542
- Lam et al. 2021, Ten questions on ANC in the built environment — https://www.sciencedirect.com/science/article/pii/S0360 (abstract plus table excerpt)
- Shen et al. 2018, MUTE (SIGCOMM) — https://sens.epfl.ch/documents/papers/sigcomm18mute.pdf
- Zhang, Pandey, Wang 2023, low-latency deep ANC — https://haozhang6720.github.io/doc/papers/ANC_journal_2023.pdf ; Deep ANC 2021 — https://pmc.ncbi.nlm.nih.gov/articles/PMC8328877/
- Silentium Quiet Bubble — https://www.silentium.com/technology-2/quiet-bubble
- Hyundai/Harman RANC — https://news.harman.com/releases/releases-20200130 ; https://www.analog.com/en/signals/ar noise-cancellation-technology.html
- Aircraft headrest virtual sensing 2023 — https://link.springer.com/article/10.1007/s42417-023-01090-5 (excerpt)
- Sound zones: https://arxiv.org/pdf/1911.10016 (excerpt); https://pub.dega-akustik.de/ICA2019/data/articles/000245.pdf (excerpt); Jeon & Choi neckband — https://pubmed.ncbi.nlm.nih.gov/33379876
- Parametric: Holosonics spec sheet — https://www.touchwindow.com/mm5/graphics/00000001/holosonics/Audio_Spotligh (excerpt); Noveto — https://www.noveto.com/licensing ; MCPL — https://arxiv.org/abs/2504.17440
- Audioscenic — https://www.audioscenic.com/ ; http://audioscenic.com/about (excerpts)
- TableConverse patent US 12,354,619 — https://patents.justia.com/patent/12354619
- Nuance Audio study — https://www.frontiersin.org/journals/audiology-and-otology/articles/10.3389/fauot.2026.1822238/f (excerpt)
- Restaurant noise and Lombard: Rindel 2012 — https://odeon.dk/pdf/C116-BNAM_2012_Rindel_29.pdf ; Hodgson 2007 — https://pubmed.ncbi.nlm.nih.gov/17471719 ; Bottalico — https://www.shs.illinois.edu/2019/07 (excerpts)
- Masking: https://pmc.ncbi.nlm.nih.gov/articles/PMC3631264/ ; https://doi.org/10.1121/1.1345696 (excerpts)
- Metamaterial: https://link.aps.org/doi/10.1103/PhysRevB.99.024302 (excerpt)

Appendix B: Hearing technology research

Compiled 2026-10-18 for the Speech Enhancement Benchmark Lab. The scope covers every form factor: open-ear glasses, neck-worn, clip-on/remote mic, tabletop, phone, distributed/hybrid. In-ear devices appear only as a comparison point.

Lab context this memo is read against: single-mic neural denoisers gave no benefit at ≤16 ms latency; a 4-mic online MVDR helped ASR; target-speaker extraction (TSE) gave +16 dB on a competing talker but runs offline.

Evidence tags: **[M]** measured (lab or clinical, numbers below). **[V]** vendor-measured (manufacturer white paper, not independent). **[C]** claimed with no published number. **[S]** speculative or research concept. **(excerpt)** means the number comes from a search-result excerpt of the cited page rather than a full read, so verify it before quoting externally.

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0. Bottom line for a founder

1. **Distance beats algorithms.** Remote or tabletop microphones near the talker give the largest measured SNR gains of any approach: **+10 to +22 dB** versus hearing-aid microphones alone. Wearable arrays give about **+2 to +10 dB**. Single-device DNN denoising adds about **+3 to +6 dB** on top of directionality in commercial products (Phonak claims up to 10 dB against an omni baseline).
2. **The latency budget is form-factor dependent.**
   - **Open-ear** devices (glasses, neckband speakers) mix processed sound with the direct acoustic path. They need about **≤5–6 ms** to avoid comb filtering (Stone & Moore 2008), and ideally <1 ms for best quality.
   - **Occluded/ANC** devices tolerate about **10–20 ms**. UW's TSH system ran at 18 ms end-to-end, and Stone & Moore 1999 put the disturbance threshold at ~20 ms.
   - **Remote-mic streaming** tolerates more when the remote signal dominates. Bluetooth-LE phone streaming arrives about 20 ms after the acoustic signal, which only works in very loud rooms.
   - Implication: our finding that single-mic DNNs fail at ≤16 ms matches the field. The products that work at low latency lean on **spatial filtering (more mics, bigger aperture)** plus a DNN post-filter, or they **move the mic closer to the mouth**.
3. **TSE is the frontier, and causal versions now exist at 8–12 ms.** UW TSH: 7.0 dB SI-SNRi at 12 ms algorithmic latency on an embedded CPU. That is well below our offline +16 dB. Expect roughly half the offline gain once causal.
4. **The most defensible product wedges** suggested by the evidence (my inference):
   - **(a) A tabletop or centerpiece array** that streams to open-ear or earbud wearables. This gets the remote-mic SNR win, can run heavy TSE on mains or larger-battery compute, and serves several listeners.
   - **(b) Glasses or a neckband with a large-aperture array** (6–16 mics), with an MVDR/binaural beamformer plus a light DNN post-filter at ≤5 ms.
   - **(c) Hybrid:** wearable array plus phone or table node, using the wireless link for look-ahead.

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1. Open-ear hearing glasses and smart glasses

Nuance Audio (EssilorLuxottica), 2025; Nuance Audio Plus, Sep 2026

- **How it works:** six microphones in the frame drive frontal beamforming plus noise reduction. Two directional open-ear speakers sit in the temples. FDA-cleared OTC hearing aid for perceived mild-to-moderate loss. It has "Frontal" and "All-around" modes.
- **Best measured benefit [M]:** Western University (Folkeard et al., *Front. Audiol. Otol.*, May 2026; n=21, mean age 75; manufacturer-funded, independent data collection):
  - Matrix SRT50 improved **2.21 dB** (−8.34 → −10.55 dB SNR).
  - CST keyword scores at −3 dB SBR rose **34%** → **63%**; at 0 dB SBR **61%** → **81%**.

  - Listening effort dropped 1.4–1.8 points on a 7-point scale.
  - The paper puts prior studies at **2.2–4.4 dB** benefit, comparable to directional open-canal hearing aids (2.6 dB).
- **Plus (2026):** still six mics and two open-ear speakers. Claims "stronger amplification, better clarity in noise," better own-voice perception, about 10 h battery. No new dB figures published [C].
- **Latency:** not published (gap).
- **Maturity:** shipping, FDA-cleared OTC.
- **Constraints:** open ear means comb filtering and limited gain. It only helps for talkers in front.
- Sources:
  - https://media.nuanceaudio.com/cms/caas/v1/media/82480/data/a970f3114db31ccb5178b7a0bfe628aa/westernuniper
  - https://www.prnewswire.com/news-releases/essilorluxottica-unveils-nuance-audio-plus-the-next-generation-of-hearing-glasses-302891793.html
  - https://hearinghealthmatters.org/hearing-technologies/2026/nuance-audio-plus-otc-hearing-glasses

Meta Ray-Ban / Oakley Meta "Conversation Focus" (announced Sep 2025; early access Dec 2025, v21)

- **How it works:** the glasses' mic array, beamforming and "real-time spatial processing" amplify the person in front. Output goes through the open-ear speakers, and the voice sounds "slightly brighter." Amplification is adjustable by swiping the temple.
- **Benefit: no published dB or latency [C].** Meta says it is not a medical device. Usage was capped at **3 h/month free** from July 2026 (UC Today report, not opened). This suggests real compute or battery cost (inference).
- **Maturity:** shipping in the US/Canada as a consumer feature.
- Sources:
  - https://about.fb.com/news/2025/12/updates-to-meta-ai-glasses-conversation-focus-spotify-integration/amp
  - https://www.meta.com/blog/connect-2025-day-one-keynote-ai-glasses-ray-ban-display-neural-band-metaverse-news/
  - https://www.uctoday.com/immersive-workplace-xr-tech/meta-ray-ban-smart-glasses-conversation-focus-limit (excerpt)

Classic eyeglass arrays: Varibel, Soede/Delft, Widrow

- **Varibel hearing glasses [M]:** a four-mic end-fire array on each temple.
  - Directional benefit **5.3 dB** in asymmetric-fit mode, **6.3 dB** in fully directional bilateral mode.
  - Anterior (temple) mics beat above-pinna mics by 2.6 dB (Boone-group study, 2010) (excerpt).
  - A patent cites a weighted directivity index of **8.2 dB** (excerpt).
- **Soede et al. 1993 (Delft) [M]:** eyeglass arrays improved SRT by about **7 dB** versus an omni mic in diffuse noise (excerpt).
- **Takeaway:** the long array aperture on a temple buys about 2–4 dB over behind-ear directionality. Nuance's measured 2.2 dB is at the low end, likely because of OTC presets, an open fit and older subjects (inference).
- Sources:
  - https://pubmed.ncbi.nlm.nih.gov/21047292
  - https://patents.google.com/patent/EP2320680A2/en
  - https://pubmed.ncbi.nlm.nih.gov/8263830

Other 2025–2026 launches

- **Cearvol (CES 2026):** Liberte open-ear hearing aid and Lyra smart hearing glasses. Claims "up to 20 dB noise reduction" for its NeuroFlow AI 2.0 platform [C, vendor; not a speech-in-noise SNR benefit]. https://hearingreview.com/hearing-products/hearing-aids/otc/cearvol-showcases-new-ai-powered-hearing-devices-at-ces-2026
- **Bone conduction:** I found no 2025–2026 bone-conduction conversation-in-noise product with measured benefit. Gap; not searched deeply.
- **Bose open-ear:** I found no Bose open-ear hearing-enhancement product. Gap.

Apple AirPods Pro 2/3: comparison point, in-ear

- **Features:** Hearing Aid feature (FDA-authorized OTC software, 2024), Conversation Boost (beam toward the talker in front), automatic Conversation Boost (iOS 26), and Live Listen (iPhone as remote mic).
- **Benefit:** Apple publishes **no dB SNR benefit or latency** in its white paper [C].
- **Note:** this is occluding plus ANC, so a different latency regime from open-ear.
- Sources:
  - https://www.apple.com/health/pdf/Hearing_Health_Features_on_AirPods_Pro_2_October_2024.pdf
  - https://support.apple.com/en-us/102479

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2. Beamforming wearables: head, chest/neck and binaural arrays

Neck/chest arrays (relevant to the neckband form factor)

- **Widrow/Luo "Hearingpoint" necklace array (2003) [M]:**
  - Six mics in a planar V on the chest, beam about 60°, link to hearing aids by telecoil neck loop.
  - HINT SNR improvement **>10 dB on average** (n=9). Word scores rose about **25% → 75%**.
  - Sidelobes sit 20–35 dB down. Feedback drops about 15 dB because the mics are far from the receivers.
  - The user aims their body at the talker.
  - https://isl.stanford.edu/~widrow/papers/j2003microphonearrays.pdf
- **Radiant Beam Array (six-mic broadfire necklace, Widrow patent) [M]:** best mean HINT improvement **7.5 dB** (speech 0°, noise 180°, sharp beam) (excerpt). https://www.audiologyonline.com/articles/clinical-results-with-six-microphone-1285
- **Etymotic/Delft assistive array (Ear & Hearing 2004):** a fixed delay-and-sum array. It is a precursor of today's table and neck arrays. https://www.phonak.com/content/dam/phonak/en/evidence-library/journal-article/Better_Speech_Perception_in_Noise_With_an_Assistive_Multimicrophone%20Array_for_Heari
- **Consumer neck devices today:**
  - BeHear PROXY: a neck speaker plus personal amplifier with no beamforming data [C].
  - Generic "neckband hearing amplifiers" with remote mics [C].
  - Bose SoundWear and Sony neck speakers are playback speakers. I found no conversation-enhancement mode or data (not verified).
  - **White space:** a modern neckband pairing a Widrow-class array (6–8 mics, chest aperture about 15–20 cm) with an MVDR+DNN and open-ear neck speakers or earbuds appears unoccupied commercially (inference).
  - https://wearandhear.com/products/behear-proxy-neck-speaker-personal-amplifier

Binaural beamformers and spatial-cue preservation

- **Binaural vs bilateral [M]:**
  - Binaural beamforming improved SRT about **6 dB**, bilateral about **4 dB**.
  - Individual range: 1.9–9.9 dB binaural, 0.3–7.4 dB bilateral (simulated classroom, HI listeners) (excerpt).
  - Commercial binaural beamformer: **5.0 dB** versus omni (excerpt).
  - Sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC8744079 ; https://www.audiologyonline.com/articles/speech-recognition-benefit-obtained-from-14338
- **Trade-off:**
  - A binaural MVDR preserves the target's ITD/ILD but collapses the cues of noise and interferers.
  - Variants that preserve cues partially (MVDR-IC, BLCMV, BMVDR-RTF) restore spatial quality with little intelligibility loss (Gössling et al. 2020) (excerpt).
  - Sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC7225838 ; https://pub.dega-akustik.de/DAGA_2026/files/upload
- **Real-time binaural DNN beamformer (GCFSnet, Westhausen et al. 2024) [M]:**
  - 4 mics (2 per ear), 2 ms frames, 168k weights, sized for a hearing-aid chip.
  - With spatial maskers, SRT80 improved **5.9 dB vs unaided** and **1.3 dB better than ADM**. It beat the binaural MVDR.
  - In diffuse cafeteria noise it was **no better than classic ADM/MVDR**.

  - Algorithmic latency 5.4 ms (14.1 ms total including lab audio hardware).
  - Assumes a zero-delay ear-to-ear link.
  - **This is close to our 4-mic MVDR finding:** a neural spatial filter wins with discrete interferers, not in diffuse babble.
  - https://arxiv.org/pdf/2405.01967

Gaze- and head-steered beamforming

- **Kidd et al. "Visually Guided Hearing Aid" (VGHA, BU, 2013–) [M, lab]:**
  - A head-band array (reported as 4×4 = 16 mics (excerpt)) steered by an eyeglass eye-tracker.
  - Beamforming benefits held up in dynamic tasks where the target moved, for both normal-hearing and hearing-impaired listeners.
  - A simulated beamformer gave 8 dB (wide) or 12 dB (narrow) noise attenuation in a gaze-vs-head study (excerpt).
  - Lab prototype only.
  - Sources: https://pubmed.ncbi.nlm.nih.gov/23464129 ; https://pubmed.ncbi.nlm.nih.gov/29252977 ; https://pmc.ncbi.nlm.nih.gov/articles/PMC6763936 ; https://www.optometrytimes.com/view/visually-guided-hearing-aid-hearing-in-plain-sight
- **Related work:**
  - Hart et al. 2009 (eye-gaze steering) is cited in the above.
  - "Cochlearity" (UC Davis 2018): an open-source eight-mic glasses array on Android with gaze steering. https://millerlab.faculty.ucdavis.edu/wp-content/uploads/sites/240/2018/07/AndersonEtAl_IEEE_2018preprint.pdf
- **Maturity [S→M]:** proven in the lab. Eye tracking in commercial glasses (Meta Ray-Ban Display, etc.) makes this newly plausible (inference).

EEG / attention-decoding ("neuro-steered") hearing aids [S]

- **CNN locus-of-attention decoding:** about **81% median accuracy with 1–2 s windows** (64-ch lab EEG). https://pmc.ncbi.nlm.nih.gov/articles/PMC8143791
- **Real-time CCA hearing-instrument control (J. Neural Eng. 2025):** 70–80% accuracy at **6–8 s** windows. https://iopscience.iop.org/article/10.1088/1741-2552/ad867c
- **Fully in-ear EEG: 61% at 60 s** windows. https://arxiv.org/html/2505.14478v1
- **Verdict:** seconds-scale switching latency and wet/dense electrodes. Years from product. Gaze or a "look once" enrollment is the practical selector today.

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3. Low-latency neural enhancement and TSE for hearables

System	Mics / platform	Benefit	Latency	Status
UW Target Speech Hearing / "Look Once to Hear" (CHI 2024)	2 binaural mics on Sony ANC headset; Orange Pi 5B CPU	**7.01 dB SI-SNRi** with noisy 5 s enrollment (clean enrollment 7.40 dB); MOS overall 2.09→3.40	8 ms chunk + 4 ms look-ahead = **12 ms algorithmic**; 6.24 ms compute; **18.24 ms end-to-end**	Research, code open [M]
Semantic Hearing (UIST 2023)	binaural, phone	**7.17 dB** average across 20 sound classes	**6.56 ms** runtime per 10 ms chunk (iPhone 11)	Research [M] (excerpt)
Sound Bubbles (Nature Electronics, Nov 2024)	up to **6 mics** on headset; embedded CPU	suppresses sources outside a 1–2 m radius by **49 dB**	8 ms chunks processed in **6.36 ms**	Research [M] (project-page excerpt)

System	Mics / platform	Benefit	Latency	Status
ClearBuds (MobiSys 2022)	2 synchronized earbuds + iPhone	separation SI-SDR gains up to ~6.9 dB in background-voices task (excerpt)	21.4 ms runtime, **109 ms end-to-end** (telephony, not live hearing)	Research [M]
Phonak Audéo Sphere Infinio "Spheric Speech Clarity" (Aug 2024)	2 mics/ear + dedicated **DEEPSONIC** DNN chip	**5.9 dB** vs open ear (default; beats competitors by 2.6–3.7 dB); **up to 10 dB** vs omni with no NR; DNN alone adds **5.8–6.9 dB** at any speech angle; 2× odds of understanding speech; tolerates 2.9 dB worse SNR at equal effort	**not published (gap)**	Shipping Rx hearing aid [V]
Phonak Roger On Sphere (remote mic, mid-Nov 2026)	Roger remote mic + DNN	**+3 dB** over the previous Roger On	"low-latency" (no figure)	Launching [V]
Oticon Intent / Own SI (DNN 2.0) (2024–25)	hearing aid	"up to 12 dB suppression of unwanted sounds"; Own SI up to **5.4 dB** SNR enhancement	n/a	Shipping [V]
Starkey Edge AI (2025)	hearing aid with DNN	brochure chart of SNR improvement at 0/−3/−6 dB babble; numbers not extracted	n/a	Shipping [V]
GCFSnet (Oldenburg 2024)	4 mics binaural	SRT80 −5.9 dB vs unaided	5.4 ms algorithmic	Research [M]
Google sub-ms SE (Dementyev et al. 2024)	min-phase FIR, low-power DSP	modest SE (see paper)	**0.32–1.25 ms algorithmic, 3.35 ms end-to-end**, 376 MIPS	Research [M] (abstract)
RT-Tango (2026)	distributed binaural DNN + SDW-MWF	competitive	8 ms, 0.49 MMAC/frame	Research (excerpt)
Causal TSE for HI listeners (2025, PMC12340209)	speaker-conditioned, causal	subjective benefit with HI listeners	low-ms (exact value not extracted)	Research

Sources: - https://arxiv.org/html/2405.06289v3 - https://tsh.cs.washington.edu/ - https://arxiv.org/pdf/2311.00320 - https://soundbubble.cs.washington.edu/ - https://www.washington.edu/news/2024/11/14/ai-headphones-sound-bubble-noise-cancelling - https://homes.cs.washington.edu/~gshyam/Papers/clearbuds.pdf - https://hearingreview.com/he products/hearing-aids/speech-in-noise/leveraging-deep-neural-networks-in-hearing-aids - https://hearingreview.com/hearing-products/amplification/assistive-devices/phonak-launches-roger-on-sphere-an-ai-powered-remote-microphone-with-real-time-speech-separation - https://www.oticon.com/professionals/campaigns-b2b/intent/intent-technology - https://assets-ue.cas.dgs.com/-/media/oticon-us/main/download-center/own-si/15500-0373-own-si-research-brief.pdf?la=en&rev=D17F - https://arxiv.org/abs/2409.18239 - https://arxiv.org/pdf/2607.01834 -

https://pmc.ncbi.nlm.nih.gov/articles/PMC12340209/

Reading against our lab results: - TSH's causal TSE gets about 7 dB at 12 ms with 2 mics. - Our offline TSE gets +16 dB. - The gap of roughly 9 dB is the price of causality and small compute. - Phonak's DNN gain is about 6 dB, but it sits on top of fixed directionality and is measured on occluded KEMAR, where latency is less constrained. - None of these validate a **single-mic, open-ear, <5 ms** DNN. That is consistent with our negative result.

Compute and power budgets: - **GreenWaves GAP9** (RISC-V, 22FDX): **330 μW/GOP**. Partner demos: AI-ANC at 1.5 mW per channel, spatial audio at 1.8 mW total. - https://www.eenewseurope.com/en/greenwaves-stresses-gap9s-hearing-credentials - https://www.eetimes.com/greenwaves-shows-off-advanced-audio-demos - **Syntiant NDP100:** <200 μW always-on (keyword-class models, not full SE). https://www.syntiant.com/press-release/syntiant-brings-speech-interfaces-to-the-edge-with-new-ultra-low-power-neural-decision-processors/ - **Back-of-envelope (inference):** - TSH's streaming TF-GridNet at about **4.6 GMAC/s** would need roughly 1.5–3 mW on GAP9 at its quoted efficiency ($\approx$9 GOP/s $\times$ 330 μW). That is plausible for glasses or a neckband, marginal for an in-ear hearing aid. - GCFSnet-class models (~0.1–0.5 MMAC/frame) fit hearing-aid chips.

Latency tolerance (perceptual): - **Open-fit: 5–6 ms** acceptable, Stone, Moore et al. 2008. https://pubmed.ncbi.nlm.nih.gov/18469715 - **Optimal quality** needs about 0.5 ms; modern hearing aids target $\leq$10 ms. https://pmc.ncbi.nlm.nih.gov/articles/PMC11638989/ - **Occluded:** disturbing beyond about 20 ms (Stone & Moore 1999). https://pubmed.ncbi.nlm.nih.gov/10386846 - **Own-voice production** is affected only beyond 30 ms (2002). https://pubmed.ncbi.nlm.nih.gov/12195175

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4. Remote-mic, tabletop, phone and distributed approaches

Remote / table microphones streaming to the listener [M]

- **Typical SNR gain:**
  - Remote mics improved SNR **16–22.7 dB** versus omni hearing aids (Rodemerk & Galster 2015, four wireless protocols) (excerpt).
  - Earlier FM work: **14.2–16.7 dB** (excerpt).
  - LE Audio literature cites 15–20 dB for direct delivery (excerpt).
  - For context, severe-to-profound loss needs +15–25 dB SNR to match normal hearing.
  - Sources: https://canadianaudiologist.ca/the-benefit-of-remote-microphones-with-four-different-wireless-protocols-feature ; https://pmc.ncbi.nlm.nih.gov/articles/PMC8072043/ ; https://spectrum.ieee.org/bluetoothl low-energy-audio
- **Phonak Roger clinical benchmark (2026) [V]:** at **−10 dB SNR**, hearing aid alone scored **12%** and HA+Roger **73%**. Benefits of +25 to +59 points across studies (excerpt). https://www.phonak.com/content/dam/phonak/en/evidence-library/field-studies/PH_FieldStudyNews_ClinicalBenchma
- **Roger Select/Pen in the table center (simulated restaurant, +5 to −10 dB SNR):** up to **61%** (Select) and **45%** (Pen) better recognition than HA/CI alone (excerpt). https://www.phonakpro.com/content/dam/phonakpro
- **Roger Table Mic (2016, n=18) [V]:** significant gains at 3.0 and 4.5 m. The fetched paper gives no dB numbers. The table mic auto-selects the active talker, and several can be networked. https://www.phonakpro.com/content/dam/phonakpro/gc_hq/en/resources/evidence/field_studies/documents/fsn_roger_
- **Adaptive gain matters:** in a bench comparison, only Roger beat a 20 cm reference mic in high noise. One competitor fell to 0% at 80 dBA (vendor study). https://www.phonak.com/content/dam/phonak/en/evidence-library/field-studies/PH_FieldStudyNews_BeyondWirelessProtocolsRogerBenchmark_210x297_EN_028-2814-02.pdf
- **Roger On Sphere (Nov 2026):** the first DNN-in-remote-mic product, +3 dB over its predecessor [V]. This validates "put the AI at the table node."

Phone as remote mic (Live Listen, apps)

- **UT Dallas SHARP-2 app (2020) [M]:**
  - iPhone on the table about 1 ft from the talker, streaming to MFi/MFA hearing aids.
  - Benefit of **+22 points** (HI) and **+29–34 points** (normal hearing) word recognition.

  - **Latency caveat:** the Bluetooth-LE stream arrives about **20 ms after** the acoustic sound. They needed circumaural headphones to mask the direct path. Real-world use works only when noise masks the early arrival.
  - https://pmc.ncbi.nlm.nih.gov/articles/PMC7744168
- **Apple Live Listen:** documented but no published latency or benefit. https://support.apple.com/en-us/102479
- **Bluetooth SIG:** echo becomes a problem beyond about 30–40 ms delay. LE Audio plus Auracast enables phone-as-mic and one-to-many broadcast (GN Multi Mic is Auracast-capable). https://www.bluetooth.com/blog/introducing-le-audio-the-use-cases/ ; https://www.gnhearing.com/en/products/auracast-hearing-aids

Distributed arrays across friends' phones / "smart table" [S→M]

- **UIC Listening Technology Lab (Corey et al., IWAENC 2022):**
  - Each diner puts their phone on the table and wears earbuds or hearing aids.
  - Adaptive crosstalk cancellation removes other diners' voices and own-voice echo from each phone.
  - Adaptive binaural spatialization re-imposes the spatial cues.
  - Demoed with real phones. **No dB in the blog.** They state **Bluetooth is too slow** for real-time listening today.
  - Follow-up work covers cooperative per-table nodes in a multi-table restaurant.
  - https://listeningtech.lab.uic.edu/news-stories/iwaenc2022-conversation ; https://listeningtech.lab.uic.edu/news-stories/cooperative-node-specific
- **Ad-hoc arrays for ASR:** Microsoft PickNet does real-time channel selection across multiple phones, with WER gains over block-online beamforming and a single distant mic. https://arxiv.org/abs/2201.09586
- **Communication-cost-aware mic selection** (Casebeer et al. 2021): streams more mics only when SNR is low. https://arxiv.org/abs/2011.07348
- **Constraints:**
  - Clock sync (ClearBuds managed <64 μs between earbuds).
  - Wireless latency and jitter.
  - Privacy and consent from the other people at the table.
  - Battery.
- **Smart table / centerpiece / lamp:** I found **no commercial product** or study with measured hearing benefit beyond Roger Table Mic-class devices. White space [S].

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5. Predictive and anticipatory ideas

- **Future-frame prediction to cut algorithmic latency [M]:**
  - Predict the next enhanced STFT frame from past frames only. With 50% overlap this gives zero algorithmic latency, leaving only buffer latency.
  - Wu & Braun (ICASSP 2025) compared this against asymmetric and learnable windows under a <5 ms budget.
  - Also used in the iNeuBe-X Clarity CEC2 entry to meet a 5 ms limit.
  - The gain is about one hop (a few ms), not tens of ms.
  - Sources: https://arxiv.org/pdf/2409.10358v1 ; https://arxiv.org/html/2302.07928v1
- **Wireless look-ahead ("faster than sound") [M for ANC, S for speech enhancement]:**
  - A remote mic radios the reference signal to the processor ahead of the acoustic wavefront (radio ≫ 343 m/s).
  - This gives N samples of look-ahead (WLANC, NTU 2024) (excerpt).
  - For a table node 1 m from a noise source and about 1 m from the listener, acoustic travel is about 3 ms per metre. That is real look-ahead budget, enough to run heavier TSE (inference).
  - https://arxiv.org/pdf/2405.12496
- **Speech prediction in ANC headphones (IEEE 2025):** frames optimal speech ANC as a non-causal prediction problem (excerpt). https://ieeexplore.ieee.org/document/11123733
- **Anticipatory gain control / traffic-noise event prediction:** I found **no published hearing-device**

work forecasting traffic or transient noise events to pre-adjust gain. Oticon Intent's head- and body-motion "4D sensors" adapt support across a **5 dB output-SNR** range by inferred intent. This is the closest commercial analogue (context-predictive, not signal-predictive) [V]. https://assets-ue.cas.dgs.com/-/media/oticon-us/main/download-center—myoticon—product-literature/intent/pi/155000264–rm–h1-2024–intent-product-guide–dc-0224v1.pdf?la=en&rev=7A1A

- **"Negative latency" by forecasting speech waveforms beyond about 1 frame [S]:** no evidence it preserves intelligibility. Speech is predictable at the phoneme level only over a few ms of waveform. Treat as speculative.

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6. Cross-form-factor comparison

Form factor	Mics / aperture	Best measured SNR benefit	Latency regime	Maturity	Key constraint
Open-ear glasses (Nuance)	6 mics, ~temple length	**+2.2 to 4.4 dB** SRT [M]	needs ≤5–6 ms (open fit)	shipping OTC	comb filter, low gain, front-only
Eyeglass end-fire arrays (Varibel, Delft)	4/temple	**+5.3 to 7 dB** [M]	same	discontinued / legacy	aesthetics, front-only
Smart glasses (Meta)	multi-mic array (count not published) + AI	no data [C]	unknown	consumer feature, time-capped	not medical
Head-band gaze-steered (VGHA)	16 mics	~8–12 dB attenuation simulated; benefits kept in dynamic tasks [M]	real-time	lab	needs eye tracker
Neck/chest array (Widrow)	6 mics, chest	**+7.5 to >10 dB** [M]	real-time (analog/telecoil)	legacy product	body-aim, link to ear
Binaural hearing-aid beamformer	4 mics	**+4 to 6 dB** [M]	<10 ms	shipping	cue loss
DNN hearing aid (Phonak Sphere)	4 + DNN chip	**+5.9 dB vs open ear; up to 10 dB vs omni** [V]	undisclosed	shipping Rx	price, power
Causal TSE headset (UW TSH)	2 + ANC	**7 dB SI-SNRi** [M]	12 ms algo / 18 ms e2e	research	needs occlusion/ANC; enrollment
Distance-based (Sound Bubble)	6 + ANC	**49 dB** outside-bubble suppression [M]	8 ms chunks	research	headset form
Remote / table mic (Roger)	1–3 near talker	**+14 to 22.7 dB** [M]	low-ms proprietary radio	shipping	talker must wear/face mic; cost
Phone on table (SHARP-2, Live Listen)	phone mics	**+22 to 34 points** words [M]	BLE ≈20 ms behind acoustic	app-level	latency vs direct sound
Distributed phones (UIC)	N phones + earbuds	demo, no dB	BT too slow today	research	sync, latency, consent

Form factor	Mics / aperture	Best measured SNR benefit	Latency regime	Maturity	Key constraint
EEG-steered	EEG + array	61–93% decode accuracy; no SNR	0.5–60 s decisions	research	electrodes

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7. Gaps and things I could not establish

- I found no published latency for Phonak Sphere, Nuance Audio/Plus, Meta Conversation Focus or Apple Conversation Boost/Live Listen.
- I found no Meta Conversation Focus SNR data.
- I found no published benefit numbers for Nuance Audio Plus.
- I found no bone-conduction or Bose/Sony neck-speaker conversation-enhancement product with data.
- I could not open the Kidd VGHA full text because PMC blocked automated access. Its array size and dB figures come from secondary excerpts.
- Several numbers are marked (excerpt): Varibel, Soede, RBA, binaural vs bilateral, remote-mic 16–22.7 dB, EEG CNN, Sound Bubble specs, Semantic Hearing, Roger clinical benchmark, WLANC. Verify them before external use.
- I found no work on traffic or transient noise-event forecasting for hearing devices.